TBO VOYAGEHACK 4.0 · MULTI SUPPLIER OPTIMIZATION

SupplyLens

Every supplier. One clear listing.

Hotel and sightseeing inventory from many suppliers, matched into one trustworthy listing per product, without losing the differences travellers care about.



Illustrative example

Team AlphaFire

Problem statement: Multi Supplier Optimization (Hotels and Sightseeing)

CONTENT 1 OF 2 · POTENTIAL IMPACT · INNOVATION

One trustworthy listing per product, with every real difference kept.

THE PROBLEM TODAY

Duplicates crowd the page

The same hotel arrives from several suppliers under different names.

Naive merging hides real differences

Pickup vs self-transfer, twin vs king, refundable vs non-refundable.

The cheapest price can be misleading

Taxes excluded, stale, or no refund at all, yet it ranks first.

HOW SUPPLYLENS SEES INVENTORY

ENTITY The same hotel or experience

3 supplier listings → Marina Pearl Hotel

↓

VARIANTS Room types, transfers, meals, access

King 32 m² = King 344 sq ft · Twin kept apart

↓

OFFERS Price, availability, cancellation

AED 1,720 with breakfast, free cancellation ✓
AED 1,580 without taxes: not comparable

POTENTIAL IMPACT

- Cleaner listing pages: one card per product
- Best bookable price surfaced: stronger price competitiveness
- Fewer post-booking surprises: higher booking conversion

INNOVATION

- Variant-safe matching, not name-only deduplication
- Best eligible offer, not lowest headline price
- A visible reason for every decision; unsure cases go to review

CONTENT 2 OF 2 · IMPLEMENTATION & TECHNICAL FEASIBILITY · ADOPTION RISK / FEASIBILITY

A simple, explainable pipeline that TBO can pilot safely.

1 Normalize

Names, units, addresses, currency. Source of every field kept.

2 Match

Same / different / needs review. Hard conflicts block merges.

3 Map variants

Rooms by occupancy, bed, size and view. Tours by inclusions.

4 Compare offers

Only complete, available prices with the required policy.

5 Rank & explain

Relevance first, then value, flexibility, freshness, reliability.

TECHNICAL FEASIBILITY

- Python + FastAPI service; no heavy infrastructure, millisecond searches
- Rules plus similarity on fields suppliers already send; no model training or GPUs
- Runs before the listing page as a sidecar to existing search

HOW WE WILL PROVE IT (PER VERTICAL)

Match precision and a harmful-merge test suite, offer-correctness checks, and ranking quality (NDCG) against raw-feed and cheapest-price baselines.

ADOPTION RISK → MITIGATION

Wrong merges	Conflicts block auto-merge; unsure pairs go to a reversible review queue
Missing or conflicting data	Shown as unknown with its source, never assumed
Stale or partial prices	Excluded from “best offer”, with the reason shown
Integration effort	Offline run on TBO sample feeds, then a controlled pilot

TEAM ALPHAFIRE · SUPPLYLENS

Thank you.

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Code & documentation: github.com/gurlivsingh/supplylens

Live demo: supplylens-alphafire.onrender.com